

Stacking Ensemble Methods

Comparative Analysis of Meta-Classifier Configurations

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Step 1: Data Preparation

1.1 Dataset Loading

- Load `dataset.mat`: 600 training, 600 test
- 2-dimensional features, binary classes (1 & 2)
- Non-linear separable configuration

1.2 Stratified Split

- Training: 420 samples (70%)
- Validation: 180 samples (30%)
- Maintains class balance

Step 2: Base Classifier Training

2.1 Model Selection

- SVM-RBF (C=10, gamma='scale')
- SVM-Polynomial (C=10, degree=2)
- Decision Tree (max_depth=None)
- Random Forest (n_estimators=100)

2.2 Hyperparameter Tuning

- GridSearchCV with 10-fold Cross Validation
- Scoring: Accuracy
- Optimal parameters per model

Table 1: Base Classifier Performance (Test Set)

Model	Accuracy	Precision	Recall	F1
SVM-Polynomial	0.9883	0.9886	0.9883	0.9883
SVM-RBF	0.9883	0.9886	0.9883	0.9883
Random Forest	0.9733	0.9734	0.9733	0.9733
Decision Tree	0.9617	0.9618	0.9617	0.9617

Step 3: Meta-Feature Generation

3.1 Predictions (Hard Labels)

- Discrete class labels (1 or 2)
- Shape: 180×4 matrix
- Model-level decisions

3.2 Scores (Soft Probabilities)

- Probability of class 2 [0,1]
- Shape: 180×4 matrix
- Confidence information

Step 4: Meta-Classifier Training

4.1 With Proper Split (Valid)

- Trained on validation meta-features
- AdaBoost with tree stumps (max_depth=1)
- n_estimators=50, learning_rate=0.1

4.2 Without Split (Control)

- Trained on training meta-features
- Same AdaBoost configuration

Table 2: Meta-Classifier Performance Comparison

Configuration	Accuracy	Precision	Recall	F1
lightgray Meta (Scores) - Split	0.9917	0.9918	0.9917	0.9917
Meta (Predictions) - Split	0.9733	0.9734	0.9733	0.9733
No-Split (Scores)	0.9717	0.9718	0.9717	0.9717
No-Split (Predictions)	0.9733	0.9734	0.9733	0.9733

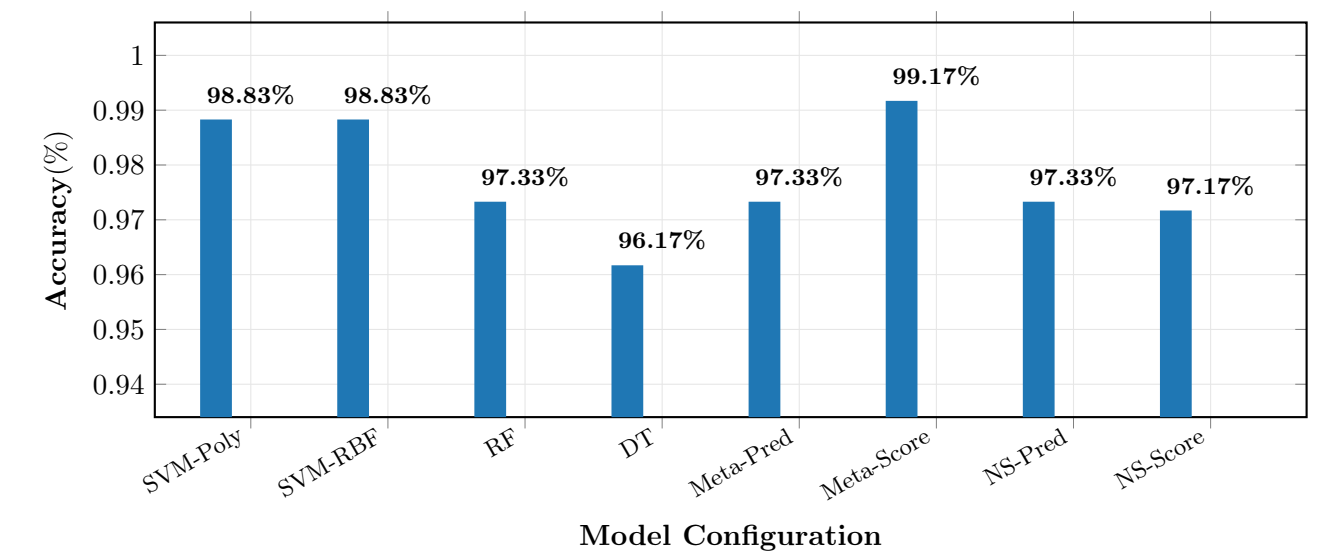


Figure 1: Comparative Performance Analysis

Step 6: Results Analysis

BEST MODEL 99.17% Meta (Scores) - Split +1.84% vs Predictions	LEAKAGE IMPACT 97.17% No-Split (Scores) -2.00% drop	BASELINE 98.83% SVM Models Best individual
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- Key Findings:
- 1. **Soft scores outperform hard predictions:** Meta-classifier using probability scores achieves **99.17%** accuracy vs. 97.33% (+1.84% improvement).
 - 2. **Proper splitting is critical:** Information leakage causes **2.00%** performance drop (99.17% → 97.17%).
 - 3. **Stacking enhances performance:** Ensemble (99.17%) outperforms best base model (98.83%) by **0.34%**.

Step 7: Conclusions

FINAL BEST MODEL Stacking Ensemble with AdaBoost meta-classifier using probability scores from SVM-RBF, SVM-Polynomial, Random Forest, and Decision Tree base models, with proper validation split.
